

Brief report

A single nucleotide polymorphism at chromosome 2q21.3 (LCT – 13910C>T) associates with clinical outcome after allogeneic hematopoietic stem cell transplantation

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A single nucleotide polymorphism (SNP) responsible for lactase persistence (LCT – 13910C>T) changes intestinal microflora. Considering the influence of bacterial microflora on various immune effects, we tested DNA from 111 recipients/donors and analyzed whether this SNP interferes with survival and the incidence of acute graft-versus-host disease (aGVHD) after allogeneic

hematopoietic stem cell transplantation (HSCT). Median overall survival (OS) was significantly longer when donors had a CC genotype (not reached after 133 vs 11.1 months, $P = .004$). Multivariate analysis identified a donor T allele (hazard ratio 2.63, 95% confidence interval 1.29–5.33, $P = .008$) as independent risk factor for death. Surprisingly, recipient geno-

types did not influence outcome and there were no differences regarding aGVHD. Transplantation-related mortality (TRM), relapse and pneumonia were significantly less frequent in patients with CC donors. These findings add to the growing list of non-HLA polymorphisms with impact on outcome after allogeneic HSCT. (Blood. 2008;112:2156-2159)

Introduction

Outcome after allogeneic HSCT is strongly influenced by immune mediated effects such as graft-versus-host disease (GVHD), graft-versus-leukemia (tumor) effects, and infectious complications. A direct influence of bacterial microflora and the gut-associated immune system on the outcome of HSCT has been suggested.¹⁻⁵ The prognostic significance of genetic NOD2/CARD15 variants⁶ and the modulation of this effect by the type of gastrointestinal decontamination⁷ further supports the important role of gut associated immunity in HSCT.

Lactose malabsorption changes the composition of colonic microflora.⁸⁻¹⁰ It can be caused by genetic polymorphisms leading to nonpersistence of lactase phlorizin hydrolase (LPH), a β -galactosidase expressed exclusively in the small intestine. In Europeans, it has been linked to an autosomal dominant polymorphism in an –14 kb upstream enhancer element of the lactase (LCT) gene¹¹ located on the long arm of chromosome 2 (2q21.3). In vitro reporter gene assays of the LCT promoter¹²⁻¹⁴ as well as mRNA transcription studies on intestinal biopsies¹⁵ indicate that this –13910C>T SNP regulates lactase transcription. For the Austrian population a frequency of approximately 80% (75%-85%) lactase-persistent individuals¹⁶ based on a –13910T-allele¹⁷ has been described.

Considering differences in the composition of intestinal microflora between lactase persistent and nonpersistent individuals we tested whether the LCT genotype interferes with the outcome of allo-HSCT and the incidence of aGVHD.

Methods

Of 114 consecutive patients who underwent first allogeneic stem cell transplantation since 1996, 111 patient/donor pairs with sufficient available

material for genotyping the LCT-13910C>T polymorphism were included in this retrospective analysis. Written consent was obtained from all patients and sibling donors as required by the institutional review board, the ethics committee of the Elisabethinen Hospital, and the Declaration of Helsinki.

Baseline patient characteristics and the distribution of standard risk factors for allogeneic HSCT are shown in Table 1.

Myeloablative conditioning regimens were mostly busulfan/cyclophosphamide (BuCy)^{18,19} or BCNU, etoposide cytarabine, cyclophosphamide, and 2-chlorodeoxyadenosine (BEAC)²⁰ without total body irradiation (TBI); reduced-intensity conditioning regimens (RIC)²¹ were predominantly fludarabine-based combinations without irradiation. Routine GVHD prophylaxis consisted of cyclosporin A (CsA) with short course methotrexate¹⁹ or combined with mycophenolate mofetil.²¹ Norfloxacin was given as standard gut decontamination.

The LCT –13910C>T polymorphism was genotyped from peripheral blood by TaqMan polymerase chain reaction (PCR). Allele frequencies (Table 1) were in Hardy Weinberg equilibrium ($P > .05$).

Categorical variables were compared by Pearson χ^2 and Fisher exact test as appropriate. Linear regression (ANOVA) was used for analysis of continuous data. OS, transplantation-related mortality (TRM), and time to relapse were analyzed with the Kaplan Meier method and the log-rank as well as Breslow test. Initial analysis for aGVHD was done for grades II-IV. Subsequent evaluation included grades I-IV, III-IV, and IV as well as organ-specific aGVHD (gut, skin, liver). Relapse or death was considered competing risks for aGVHD. Multivariable analysis was performed using Cox-regression analysis. Factors found to be significant in univariate analyses ($\alpha \leq .2$) were included in the model. These include stage of disease and donor LCT genotype for all outcomes, source of stem cells (bone marrow or peripheral blood) for OS, and type of donor (HLA identical sibling vs matched unrelated donor or HLA mismatched donor) for TRM and relapse. P values were 2-sided and

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Table 1. Patient characteristics and distribution of common risk factors

	All patients	Donor LCT –13910 CC	Donor LCT –13910 TC or TT	P
Number in group (%)	111	30 (27.0)	81 (73.0)	
Patients' mean age, y, (± SD)	41.7 (± 12.5)	44.4 (± 11.3)	40.7 (± 12.8)	.170
Younger than 40 y, n (%)	50 (45.0)	10 (33.3)	40 (49.4)	.141
Older than 40 y, n (%)	61 (55.0)	20 (66.7)	41 (50.6)	
Sex, n (%)	.0			
Male	60 (54.1)	18 (60.0)	42 (51.9)	.444
Female	51 (45.9)	12 (40.0)	39 (48.1)	
Conditioning regimen, n (%)				
Myeloablative	70 (63.1)	17 (56.7)	53 (65.4)	.395
Reduced intensity	41 (36.9)	13 (43.3)	28 (34.6)	
Source of stem cells, n (%)				
Bone marrow	8 (7.2)	2 (6.7)	6 (7.4)	1.000
Peripheral blood stem cells	103 (92.8)	28 (93.3)	75 (92.6)	
Stage of disease, n (%)				
Early*	41 (36.9)	11 (36.7)	30 (37.0)	.971
Advanced†	70 (63.1)	19 (63.3)	51 (63.0)	
Diagnosis, n (%)				
Acute leukemia	69 (62.2)	20 (66.7)	49 (60.5)	.621
Chronic myeloid disorder	20 (18.0)	5 (16.7)	15 (18.5)	
Lymphoma	14 (12.6)	2 (6.7)	12 (14.8)	
Other	8 (7.2)	3 (10.0)	5 (6.2)	
Type of donor, n (%)				
HLA identical sibling	87 (78.4)	20 (66.7)	67 (82.7)	.068
Other donor (mismatch and/or UD)	24 (21.6)	10 (33.3)	14 (17.3)	
Donors' mean age, y (± SD)	40.9 (± 12.9)	40.0 (± 14.3)	41.3 (± 12.5)	.637
Younger than 40 y, n (%)	51 (45.9)	14 (46.7)	37 (45.7)	.926
Older than 40 y, n (%)	60 (54.1)	16 (53.3)	44 (54.3)	
Donor/recipient sex, n (%)				
Female/male	21 (18.9)	5 (16.7)	16 (19.8)	.712
Other combination	90 (81.1)	25 (83.3)	65 (80.2)	
Recipient/donor CMV status, n (%)‡				
Positive/negative	17 (16.5)	6 (20.7)	11 (14.9)	.474
Other combination	86 (83.5)	23 (79.3)	63 (85.1)	
CD 34 ⁺ cells infused, 10 ⁶ /kg (± SD)	4.7 (± 2.3)	5.0 (± 2.4)	4.5 (± 2.3)	.362
Recipient LCT –13910 genotype, n (%)				
TT	32 (28.8)	4 (13.3)	28 (34.6)	<.001
TC	59 (53.2)	13 (43.3)	46 (56.8)	
CC	20 (18.0)	13 (43.3)	7 (8.6)	

*Includes acute leukemia in first complete remission, chronic myeloid leukemia in first chronic phase, and severe aplastic anemia.

†All other states.

‡CMV serostatus is unknown for 5 donor/recipient pairs (all with TC or TT donors).

outcomes were considered to be significant with an α level of less than or equal to .05.

Results and discussion

Characteristics of patients in the 2 groups were generally well matchable with regard to common risk factors (Table 1). However, there were nonsignificant trends toward higher age and fewer HLA-identical sibling transplants in the group with a CC genotype donor. There is an uneven distribution of donors and recipients genotypes. Thirteen of 20 patients with a CC genotype had a CC donor ($P < .001$). This imbalance is mainly caused by nonrandom donor selection due to sibling donors.

The median time of follow-up in surviving patients was 25.8 months (range 0.26 to 133.1). OS was significantly longer for patients with an LCT –13 910 CC donor than for those with a TC or TT genotype donor (log-rank $P = .004$, Breslow $P = .006$). Median survival of the latter was 11.1 months while it was not reached with a CC donor (Figure 1A). A plateau in OS was seen at

61.0% with CC donors, while it was at only 26.9% when the donor carried a T allele. This survival benefit was seen in all subgroups analyzed and reached a statistical significant difference in patients over 40 years, AML patients, advanced stage patients, HLA identical sibling transplants and myeloablative conditioning (Figure S1, available on the *Blood* website; see the Supplemental Materials link at the top of the online article). Multivariable analyses identified advanced stage (hazard ratio [HR] 2.57, 95% confidence interval [CI] 1.42-4.64, $P = .002$) and a donor T allele (HR 2.63, 95% CI 1.29-5.33, $P = .008$) as independent risk factors for shorter survival.

No significant difference in OS was seen according to recipients' LCT –13910 genotype.

No relevant difference in the incidence or severity of aGVHD with regard to recipients' or donors' LCT genotype could be found. Organ involvement of aGVHD was not associated with LCT genotypes.

Because one of the primary end points of this retrospective analysis (OS) clearly resulted in a significant difference, additional analyses were done for TRM and relapse or progression.

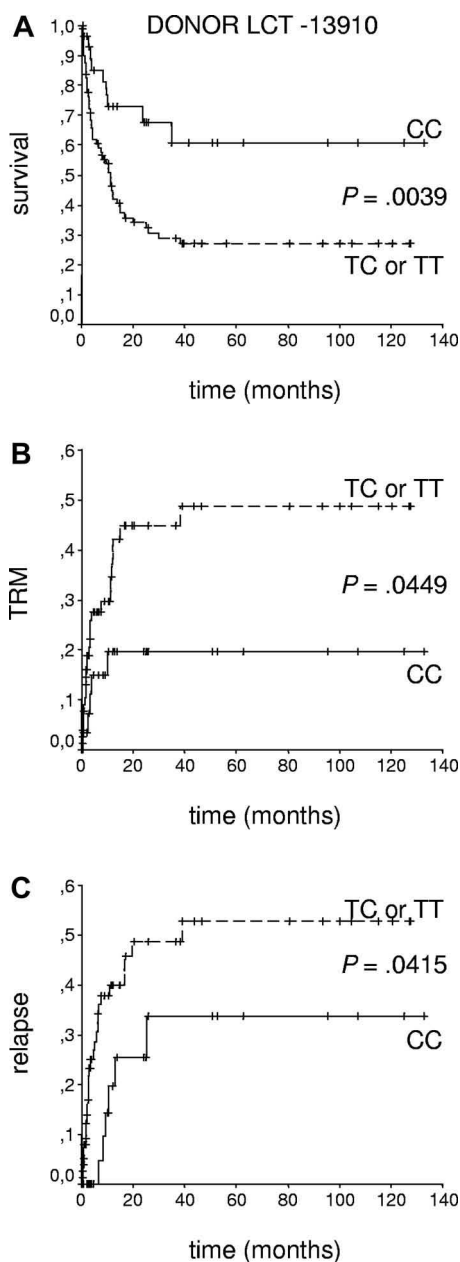


Figure 1. Overall survival, TRM, and relapse or progression according to donor LCT-13910C>T genotypes. Patient numbers: CC: n = 30; TC or TT: n = 81. (A) Kaplan-Meier analysis of overall survival, (B) cumulative incidence of TRM, (C) cumulative incidence of relapse or progression.

In patients with a CC donor a significantly lower incidence of TRM, defined as death from any cause in nonrelapsed patients,²² was seen with the log-rank test (Figure 1B; log-rank $P = .045$, Breslow $P = .062$).

Similarly, a significant difference regarding the probability of relapse or progression depending on the donor LCT-13910C>T SNP was observed (Figure 1C). Median time to relapse or progression was 38.9 months for patients whose donor carried a T allele, while it was not reached in patients with CC donors (log-rank $P = .042$, Breslow $P = .012$).

Cox multivariable analysis showed advanced stage of disease (HR 3.20, 95% CI 1.47-6.96, $P = .003$), unrelated or mismatched donor (HR 3.69, 95% CI 1.72-7.91, $P < .001$) and any T allele at the donor's LCT-13910 locus (HR 3.79, 95% CI 1.48-9.72, $P = .006$) as significant risk factors for relapse.

On the contrary, recipients' LCT genotype did not result in any significant differences for TRM and relapse.

Posttransplantation pneumonia was significantly less frequent in recipients of grafts with a CC genotype. Until day 100 after HSCT, 28.6% of patients with a TC or TT donor versus only 3.8% of patients with a CC donor suffered from pneumonia ($P = .009$). No significant association was seen between LCT genotypes and the incidence of other infectious complications such as septicemia, fever, CMV reactivation, or other viral infections.

However, a survival benefit was associated with donor but not recipient genotypes. Thus, these results do not support our original expectations. Our data suggest that the observed difference in survival is mainly associated with the genotype of transplanted cells. Whether this points toward immune-mediated effects remains to be determined in further studies. The T allele of the LCT polymorphism lies within a large haplotype extending more than 1 Mb at the long arm of chromosome 2 (2q21.3) due to recent²³ and strong evolutionary selection of the LCT-13910 T allele. Because of this high degree of linkage disequilibrium,^{24,25} the observed difference in survival is not necessarily related to the LCT gene directly. However, although this SNP is a reasonably good surrogate marker for the whole genomic region, it would be difficult for genetic association studies to assign the observed effects to another nearby gene and, therefore, another SNP.

These findings add to the growing list of non-HLA polymorphisms with impact on outcome after allogeneic HSCT and should be evaluated in an independent larger series.

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Authorship

Contribution: H.H. wrote the manuscript, performed statistical analyses, and interpreted data; O.Z. performed PCRs; O.K., H.K., J.K. and M.G. treated patients and collected clinical data and patients materials; R.O. supervised statistical analyses and reviewed the manuscript; and D.L. designed and supervised the project and critically reviewed the manuscript.

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